

7. In a diffusion experiment, gelatin was mixed with a pink pH indicator and allowed to set. The gelatin was then cut into different sized cubes. The cubes were placed in a beaker containing dilute acid. As the acid diffused through the gelatin the cubes became colourless. The time taken to change from pink to colourless was recorded.

The surface area and volume of the cubes were calculated using the following equations:

$$\text{surface area of cube} = \text{length} \times \text{width} \times \text{number of sides}$$

$$\text{volume of cube} = \text{length} \times \text{width} \times \text{height}$$

- (a) The results of the experiment are shown below.

Length of side of gelatin cube (cm)	Time taken to become colourless (s)	Surface area of cube (cm ²)	Volume of cube (cm ³)	Surface area:volume ratio
1	20	6	1	6:1
5	41	150		6:5
7	104		343	6:7
10	610	600	1000	

- (i) **Complete** the table. [3]
- (ii) Sarah suggested that diffusion of the acid into the gelatin is quicker in the smaller cubes because the time taken for them to become colourless is less. Explain whether you agree with Sarah. [2]

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- (b) Red blood cells transport oxygen around the body to allow respiration to occur in the tissues. Red blood cells have a volume of 90 units and a surface area of 136 units. Explain how the surface area:volume ratio of red blood cells make them suitable for their function. [3]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1										4 He Helium 2																																		
7 Li Lithium 3 9 Be Beryllium 4 23 Na Sodium 11 24 Mg Magnesium 12 39 K Potassium 19 40 Ca Calcium 20 86 Rb Rubidium 37 88 Sr Strontium 38 133 Cs Caesium 55 137 Ba Barium 56 223 Fr Francium 87 226 Ra Radium 88 227 Ac Actinium 89 227 Ac Actinium 89															11 B Boron 5 12 C Carbon 6 14 N Nitrogen 7 16 O Oxygen 8 19 F Fluorine 9 20 Ne Neon 10					27 Al Aluminium 13 28 Si Silicon 14 31 P Phosphorus 15 32 S Sulfur 16 35.5 Cl Chlorine 17 40 Ar Argon 18					45 Sc Scandium 21 48 Ti Titanium 22 51 V Vanadium 23 52 Cr Chromium 24 55 Mn Manganese 25 56 Fe Iron 26 59 Co Cobalt 27 59 Ni Nickel 28 63.5 Cu Copper 29 65 Zn Zinc 30					70 Ga Gallium 31 73 Ge Germanium 32 75 As Arsenic 33 79 Se Selenium 34 80 Br Bromine 35 84 Kr Krypton 36					115 In Indium 49 119 Sn Tin 50 122 Sb Antimony 51 128 Te Tellurium 52 127 I Iodine 53 131 Xe Xenon 54					204 Tl Thallium 81 207 Pb Lead 82 209 Bi Bismuth 83 210 Po Polonium 84 210 At Astatine 85 222 Rn Radon 86				
Key																																												

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number